

How to store data on DNA?

DNA 数据存储（字幕版）｜如何把信息写进生命分子，从二进制到碱基序列

时长 11:50 | 句数 173 | 术语 1 | 章节 7

摘要

本讲共 7 章，从「录像是应下列记者的要求制作的」到「实例」。全文 173 句、173 句已译成中文，标注了 1 个 AI 术语（主要集中在神经网络基础）。

7 chapters, 173 sentences (173 translated), 1 AI terms annotated. Built fully offline with opus-mt + local glossary.

录像是应下列记者的要求制作的 00:00

00:00 This video was produced at the request of
录像是应下列记者的要求制作的:

00:02 EcoScience Sud Provence Albe Code des Ur and powered by Player's Ur Prod.
由玩家的Ur Prod提供动力

00:06 It was co-scripted with Marc Antonini,
上面是马克·安东尼尼的专栏

00:08 CNRS Research Director at the I3S Laboratory,
国家科学研究中心I3S实验室研究主任,

00:11 a CNRS and University of Code des Ur Laboratory in France.
a 法国国家遥感中心和乌尔大学实验室守则。

00:21 Welcome back to ScienceClick.
欢迎回到科学闪烁。

00:23 Today, what can we store in DNA?
今天,我们有什么可以储存在DNA里?

00:26 With the rise of computing,
随着计算的增长,

我们的社会越来越依赖数据存储 00:30

00:30 our society has come to rely more heavily on data storage,
我们的社会越来越依赖数据存储

00:33 photos, videos, music, text.
照片,视频,音乐,文字

00:36 All these files, kept on our personal computers and smartphones,
所有这些档案 都放在我们的个人电脑和智能手机上

00:40 but mostly in data centers, represent several trillion gigabytes,
但大部分在数据中心 代表数万亿千兆字节

00:44 a quantity that continues to grow
a 数量继续增加

00:46 and which would take billions of years to download.
下载需要数十亿年

00:50 If we stored them on DVDs,
如果我们把它们储存在DVD上

00:52 they would form a pile stretching far higher than the moon.
将形成一个堆积,其伸展的比月亮还要高。

00:56 To store all these files,
为了存储所有这些文件,

00:58 data centers consume colossal amounts of resources and energy.
数据中心消耗了大量资源和能源

01:02 They require huge buildings,
他们需要巨大的建筑,

01:04 amounts of billions of litres of water for cooling
数十亿升水供冷却

01:07 and can have a carbon footprint equivalent to that of an entire city.
并且可以拥有相当于整个城市的碳足迹。

01:10 Digital infrastructure as a whole
整个数字基础设施

01:12 emits approximately as much greenhouse gas as all air traffic.
排放的温室气体大约与所有空中交通一样多。

01:16 It is estimated that 60-80% of all stored files are virtually never read or used.
据估计,所有储存的档案中,60-80%实际上从未读或使用过。

01:24 We call this cold data,
我们称之为冷数据

01:26 which is stored away without anyone accessing it.
却被存放在无人可及的地方。

01:29 It encompasses archives, old emails, legal or administrative documents,
它包括档案、旧电子邮件、法律或行政文件,

01:33 files that data centers must conserve all the same.
数据中心必须保存所有文件。

01:37 Today, we are faced with an exponential increase in such stored data.
今天,我们面临着此类储存数据急剧增加的问题。

01:43 For several decades,
几十年来,

研究人员一直想知道是否可以储 01:44

01:44 researchers have been wondering whether this cold data could be stored
研究人员一直想知道是否可以储存这种冷数据

01:47 in a more passive and efficient way.
以更被动和高效的方式。

01:50 One idea proposed in the 1950s by Richard Feynman
1950年代理查德·费曼提出的一个想法

01:53 would be to take inspiration from living beings.
将受到生命的启发

02:03 Indeed, all living beings have tiny storage spaces.
事实上,所有生物都有很小的储存空间。

02:07 In almost every cell of our body is kept our genome,
在我们身体的每一个细胞里 几乎每个细胞都保留着我们的基因组

02:11 a kind of file that contains all the instructions necessary for our development,
一种包含我们开发所需的所有指示的文件,

02:16 such as the colour of our eyes or our blood type.
诸如我们眼睛的颜色或血型。

02:19 For a human being, this file amounts roughly to 1 GB of information,
对于一个人来说,这个档案大约相当于1GB的信息

02:23 the equivalent of a movie in standard quality.
相当于标准质量的电影。

02:26 It is stored physically in the form of a molecule,
它以分子的形式物理储存

02:29 a collection of atoms, DNA.
收集原子和DNA

02:33 DNA is essentially a 2-metre-long string
DNA基本上是一个2米长的字符串

02:36 made up of smaller molecules, nucleobases,
由小分子、核基组成

02:39 which exist in four versions, A, T, G and C,
现有四种版本,A、T、G和C,

02:43 and which form a sort of alphabet in which our genome is written.
形成一种字母, 我们的基因组在其中写作。

02:47 In our cells, DNA is curled up on itself
在我们的细胞里 DNA是自己凝固起来的

02:51 and contained in a tiny nucleus of a few micrometres,
以若干微米的极小核盟誓,

02:55 a storage 100,000 times more compact than the traditional SD card.
存储量比传统的SD卡的压强10万倍

03:01 So could we take inspiration from DNA to store our files more efficiently?
所以我们能不能从DNA中得到灵感 来更高效地存储我们的档案?

03:07 Since 2013, it is now possible to store information, images
自2013年以来,现在可以储存信息、图像

03:12 or any other type of file on DNA.
或DNA上的任何其他类型的文件。

03:15 In this video, we will see together how this works.
这部影片将让我们一起看这个过程如何运作。

03:27 Put simply, a computer file is a very large number.
简而言之,计算机文件是一个非常大的数字。

实例 03:31

03:31 To store text, for instance, we code each letter and character
例如,要存储文字,我们编码每个字母和字符

instance ['instəns]
实例 与样本 的含义相同。
神经网络基础

03:35 with a number and assemble these numbers in a sequence.
与数字并列,然后按顺序组装这些数字。

03:38 To store an image, the amount of red, green and blue in each pixel
存储图像, 每个像素中红色、绿色和蓝色的量

03:43 is quantified and these numbers are put one after the other, pixel after pixel.
将量化,然后将这些数字相继排列,像素等值与像素等量相继。

03:49 Storing a file thus amounts to physically encoding a number.
因此,存储文件等于将数字编码为物理编码。

03:54 The first step is encoding.
第一步是编码。

03:57 Computers use electricity.
计算机使用电力。

04:00 The voltage may be either low or high
电压可能低,也可能高

04:02 and can thus be used to represent two digits, zero and one.
并可用于代表两位数,零位数和一位数。

04:07 These two digits are enough to encode any number using the binary counting system
这两个数字足以用二进制计数系统编码任何数字

04:12 and thus store any file.
保存任何文件。

04:15 The number nine, for instance, is written in binary 1001.
例如,九号是用二进制1001书写的。

instance ['instəns]
实例 与样本 的含义相同。
神经网络基础

04:23 On the other hand, DNA is a sequence of four bases, A, T, G and C,
另一方面,DNA是四个基点的序列,A、T、G和C

04:29 which can be interpreted as four digits, 0, 1, 2, 3.
4位数,0,1,2,3

04:33 We call this quaternary.
我们称它为四月。

04:36 An advantage of quaternary is that it uses half as many digits as binary
甲长的优势在于它使用二进制的一半数字

04:40 to represent the same number.
表示相同数字。

04:42 In quaternary, the number nine, for instance, is written 21.
以四点为例,九点是21点

instance ['instəns]
实例 与样本 的含义相同。
神经网络基础

04:48 To store a file in the form of DNA, we must therefore start by translating
因此,要以DNA形式存储文件,我们必须首先翻译

04:52 from the language of computers, binary, to the language of DNA, quaternary.
从计算机的语言,二进制, 到DNA的语言,四进制。

04:58 In theory, it is a mere conversion of numbers,
理论上,这只是数字转换而已

05:01 but it is necessary to take into account biochemical constraints,
但是必须考虑到生化方面的制约因素,

05:04 given the fact that we want to create a molecule
以我们想要创建分子的事实为条件

05:07 and find an encoding method that is robust to noise and errors.
并找到一种对噪音和错误 坚固的编码方法。

05:11 After encoding, the second step will be to create the DNA molecule
编码后, 第二步将是创建DNA分子

05:16 to store our file. This is called synthesis.
保存我们的文件。 这叫做合成。

05:20 This DNA is inspired by biology but has no biological use.
DNA是生物学启发的,但没有生物用途。

05:25 It is just a physical object that we build to hold our data.
这只是我们为保存数据而建造的物理物体。

05:29 Thanks to molecular biology, this is possible,
多亏分子生物学 这有可能

05:32 and researchers are able to synthesize DNA molecules.
研究人员能够合成DNA分子。

05:36 In practice, we cut the file into several shorter strands
在实践中,我们把档案分成几个短的线条

05:41 and synthesize multiple copies of these strands
并合成这些线条的多份副本

05:44 to cross-check information and avoid errors.
以交叉核对信息和避免错误。

05:51 Finally, the last step is to store our DNA strands.
最后,最后一步是 储存我们的DNA片段

05:55 At first glance, the molecules that we synthesized
初一眼,我们合成的分子

05:58 already contain our file, but many external factors
已经包含我们的文档, 但有许多外部因素

06:02 risk degrading them. It is important to isolate the DNA
将DNA分离出来很重要。

06:06 from oxygen, humidity, light or heat, which could make it
氧、湿度、光或热能,可以使它

06:10 unstable, introduce errors or even break up the molecules.
不稳定,引入错误 甚至分裂分子。

06:15 To do this, we synthesize dehydrated DNA, which is then
为此,我们合成脱水DNA

放在一个从外表保护的小型密封 06:19

06:19 placed inside a small airtight metal capsule protected from the outside.
放在一个从外表保护的小型密封金属胶囊内。

06:24 Once stored, this DNA can remain readable for thousands of years.
一旦储存,这种DNA可以几千年保持可读性。

06:30 The molecules in the capsule are so small that it can contain
囊中分子太小,足以容纳

06:34 billions of strands and in theory store the contents of an entire data center.
数十亿条线,理论上储存整个数据中心的内容。

06:39 100 million times denser than a standard hard drive,
比标准硬盘密度高一亿倍

06:43 the equivalent of holding all the data in the world inside a shoebox.
等于将世界上所有数据都保存在鞋盒内。

06:49 For the method to be useful, it is necessary to be able to
为使方法有用,必须能够

06:52 extract files from this DNA if we wish to read them.
如果我们想阅读,请从这个DNA中提取文件。

06:56 Since the 1970s, researchers have developed technologies to read DNA.
自1970年代以来,研究人员开发了阅读DNA的技术。

07:01 This is called sequencing.
这叫测序

07:04 These techniques are generally used in biology.
这些技术通常用于生物学。

07:08 Starting from DNA molecules, they allow us to find the
从DNA分子开始, 他们让我们找到

07:11 sequences of nucleobases which compose them.
形成这些细胞的核核基序列

07:15 Researchers recently managed to read the DNA of a million-year-old mammoth.
研究人员最近读到了一百万年长的猛毛象的DNA

07:20 The same process can be used to read artificial DNA
同一程序可用于阅读人工DNA

07:25 and then by converting it back into binary, find the file that was encoded in it.
然后把它转换成二进制, 找到其中编码的文件。

07:31 Storing files on synthetic DNA has many advantages.
合成DNA的储存档案有许多好处。

07:42 Extremely compact, very stable over time, it is a passive storage solution
极紧凑,长期稳定,是一种被动储存办法

不需要任何精力来维持 07:47

07:47 which requires no energy to maintain.
不需要任何精力来维持

07:51 However, if this technology is not yet widespread,
然而,如果这种技术尚未普及,

07:54 it is because it also has certain disadvantages.
这是因为它也有一些不利之处。

07:57 When reading a file, the sequencing process introduces errors
当阅读文件时, 排序程序会引入错误

08:01 so it is currently necessary to store multiple copies of the same DNA strands
因此目前有必要储存相同DNA条的多份副本

08:06 to cross-check information and avoid errors.
以交叉核对信息和避免错误。

08:09 Another major issue is speed.
另一个主要问题是速度。

08:13 Synthesizing or sequencing molecules takes far too long.
合成或测序分子需要的时间太长。

08:17 In 2024, it takes two minutes to write a single nuclear base,
2024年,写一个单一的核基地需要两分钟

08:22 billions of times slower than a traditional hard drive.
比传统的硬盘驱动器慢数十亿倍

08:25 The technology is also very expensive,
科技也非常昂贵

08:28 around $1,000 per megabyte of synthesized DNA.
每兆字节约1,000美元 合成DNA。

08:32 Despite this, researchers are optimistic.
尽管如此,研究人员还是乐观。

08:35 We can already parallelize the synthesis process
我们可以已经可以同时开展综合进程

08:38 and achieve writing speeds of around one kilobyte per second.
并达到大约每秒1千字节的写作速度。

08:41 By reducing the margins of synthesis companies
减少合成公司的利润幅度

08:44 as well as optimizing the whole process,
以及优化整个进程,

08:46 it should be possible to reach $1 per gigabyte within a few years
几年内每千兆字节应可达到1美元

08:50 and a speed roughly that of an internet connection.
和互联网连接的速度。

08:54 In any case, at present, the end goal of this technology
无论如何,目前这一技术的最终目标

实例 08:57

08:57 would only be to store cold data,
只会存储冷数据

09:00 archives that must be preserved
必须保存的档案

09:02 without having to be consulted regularly.
无需定期咨询。

09:08 At the Computing, Signals and Systems Laboratory of Sophia Antipoulos,
在索菲亚安提普洛的计算机、信号和系统实验室

09:12 a CNRS and University of Côte d'Azur laboratory in France,
a 法国的CNRS和Côte d'Azur大学实验室,

09:15 Marc Antonini is a CNRS research director Marc Antonini是CNRS研究主任
Marc Antonini是CNRS研究主任Marc AntonnieterCNRS 马克·安东尼尼埃玛·CNRS

09:18 and Laureate of the 2023 CNRS Innovation Medal.
2023年CNRS创新奖章获得者。

09:22 He directs the Exploratory PEPR Molecule Archive,
他领导探索性PEPR分子档案馆

09:26 a research program dedicated to molecular data storage
专门研究分子数据存储的研究方案

09:29 and aimed precisely at developing these techniques.
其目标正是发展这些技术。

09:32 Launched in 2022 and funded by France 2030,
2022年启动,法国出资,法国2030年

09:36 this program aims to explore various storage and sequencing technologies
该方案旨在探索各种储存和排序技术

09:40 and accelerate synthesis methods to adapt existing systems to this new usage.
并加快综合方法,使现有系统适应这一新用途。

09:46 Many topics are explored.
探讨了许多专题。

09:48 For instance, it is important to define standards
例如,必须界定标准

instance ['instəns]
实例 与样本 的含义相同。
神经网络基础

09:51 for structuring the data stored on the multiple strands of DNA.
用于构建储存在DNA多条线上的数据结构。

09:55 If a capsule contains billions,
如果胶囊含有数十亿

09:57 we must be able to quickly find the file we're interested in.
我们必须能够迅速找到我们感兴趣的文件

10:01 It is also necessary to consider strategies for compressing files
还需要考虑压缩档案的战略

10:05 and especially images which constitute a large portion of all stored data.
特别是占所有存储数据大部分的图像。

10:11 DNA takes a long time to synthesize
DNA需要很长时间合成

10:13 and compressing file size therefore remains crucial.
因此压缩文件大小仍然至关重要。

10:17 At the I3S laboratory, Marc Antonini leads a team on this subject.
在I3S实验室,马克·安东尼尼领导了一个小组处理这个课题。

10:22 He is president of the international group JPEG DNA,
他是JPEGDNA国际集团的主席

10:26 which aims at defining an image compression standard specific to DNA by 2025.
目的是在2025年之前确定DNA特有的图像压缩标准。

10:32 Several of these projects are carried out in collaboration with actors in the field of archiving,
其中一些项目是与档案领域行为者合作开展的,

10:39 such as the National Audiovisual Institute, the European Parliament,
例如,国家视听研究所、欧洲议会、欧洲议会、美国国家视听学院、欧洲议会等。

10:43 the French National Library or Software Heritage.
法国国家图书馆或软件遗产。

10:46 They try to set up real large-scale testing.
他们试图建立真正的大规模测试。

10:49 DNA data storage has become a research avenue
DNA数据储存已成为研究渠道

10:56 at the crossroads of multiple disciplines,
位于多种学科交汇处,

10:59 microfluidics, signal theory, bioinformatics, genetic sequencing and polymer chemistry.
微氟化物、信号理论、生物信息学、基因测序和聚合化学。

11:05 Several researchers and companies actively look into parallelization methods
若干研究人员和公司积极研究平行方法

11:10 to speed up the synthesis process.
以加快合成过程。

11:13 Others are interested in how a file stored on DNA could be secured.
其他人则对如何确保DNA储存的文件安全感兴趣。

11:18 Companies are already using artificial DNA as a marker to trace stolen items.
公司已经在使用人工DNA作为标记,以追踪被盗物品。

11:25 Finally, some researchers are even interested in creating DNA computers,
最后,一些研究人员甚至有兴趣 创建DNA电脑

11:29 machines that could act at the molecular level to carry out calculations or solve problems,
可以在分子一级采取行动的机器 进行计算或解决问题,

11:34 and in which electronic chips give way to enzymes and strands of DNA.
电子芯片让位给酶和DNA的链条

11:39 Since the 1990s results have been obtained,
自1990年代以来,取得了以下成果:

11:43 and in 2003, three researchers even managed to create the very first game of DNA tic-tac-toe.

2003年,三位研究人员甚至设法创造了第一场DNA 滴答答答的游戏。

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